Temporal Symptom Extraction from Doctor-Patient Dialogues

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I. Introduction

Information extraction from electronic health records (EHR) data is an ever-growing area of study due to its potential to improve healthcare and doctor-patient interactions. In particular, conversations between doctors and patients contain valuable information regarding current and past patient symptoms and medical conditions. Thus, extracting symptoms and their respective temporal information is vital in determining the urgency of symptoms and provide better support for automated systems at the point of care. This new method can aid in the creation of more effective clinical pathways and building an accessible patient-symptom timeline. In addition, methods to extract information from dialogue in this way can be applied across many fields where dialogue of this nature occurs and is not limited to healthcare alone.

This extraction method is, however, challenging due to its novelty and the casual grammatical nature of dialogues. Though many symptom extraction methods exist, to extract temporal information for these symptoms our method will need to parse through the various verb cases and grammatical structures in which a patient can express experiencing a symptom. This includes vague time references (ex. "I have been having a cough" vs. "I have a cough") and informal language (ex. "My throat feels funny and scratchy" vs "I have a sore throat"). Additionally, there are no pre-existing annotated data sources for current and past symptoms in a doctor-patient dialogue, so we will need to create these annotations ourselves. Finally, though temporal extraction methods already exist for clinical texts, they depend on tagging methods that recognize more explicit mentions of dates and times, and have not been used on dialogue transcriptions. Thus, we must create our own temporal extraction method to combine with symptom extraction to achieve our goal of temporal symptom extraction from dialogue texts.

We propose a two-part extraction method that first uses a $\text{CharacterBERT}_{\text{medical}}$ model trained on MIMIC-III data to extract patient symptoms from dialogue texts, marking the symptom words as 'problem' and non-symptom words as '0'. Next, these 'problem' words are passed to a rule-based classifier, we examine the dependency parse tree of the context surrounding the word (ie. the remaining sentence) using the Stanford Stanza library's default DepparseProcessor. This will then assign the 'problem' word (i.e the symptom) with either 'past' or 'current'. The overall output will be extracted symptoms (ex: 'cough') and their temporal context (either 'past' or 'current').

II. Related Works

Symptom extraction from medical text has been a huge focus in NLP research. Several studies have successfully extracted patient symptoms from a variety of clinical texts, including Electronic medical notes (See Divita, G et al. 2017) and patient records (See Jackson, RG et al. 2017). Additionally, the extraction of temporal information is another focus within NLP research. For example, the TIE algorithm (See Ling, X.; Weld, D. 2010), was a novel algorithm that uses a probabilistic model to recognize temporal relations between events in a sentence through the recognition of temporal expressions.

Additionally, MEDFILTER (See Kholsa, S et al. 2020) is a model which, when given a transcribed clinical conversation, classifies and outputs the medically relevant utterances in the conversations using a BERT based encoder and a Bidirectional Long short-term memory network. The researchers in this study noted the challenges of speaker roles and knowledge in a bi-directional conversation, as well as general conversation noise, such as false starts and topic diffusion. In addition, as a conversation progresses, much of the shared information becomes implicit due to the continuous building of context, making it difficult to determine relationships between certain events mentioned in the conversation.

Finally, one study focused on extracting both temporal and symptomatic information from EHR records to determine how long a patient had an undiagnosed mental health disorder (See Viani, N. et al. 2020) In this study, a rule-based NLP system was used for the time expression extraction. The tasks of symptom extraction and time expression extraction were divided, and then a temporal "linking" method was used to combine their results. The temporal extraction piece used pre-existing tagging methods, such as SUtime and HeidelTime, to tag time expressions in the text. These time expressions include dates (ex "2010-03-01"), age related expressions (ex "when the patient was 10") and relative time expressions ("3 days ago").

Our work advances the technical community as it combines both symptom extraction and temporal extraction on an underutilized piece of clinical text: doctor-patient dialogue. Unlike the affirmation works, our method allows for accurate temporal symptom extraction without the need for explicit temporal expressions, such as dates or quantitative time frames (ex. "3 days ago"). This allows for both doctors and automated symptoms to recognize the patient's chief complaints even if the patient doesn't specify the date or time-frame of the symptom. Furthermore, it enables the construction of a timeline of current and past symptoms which can be used to develop a clinical pathway for the patient without the need for a formal patient write up. It is also important to note that due to the adaptability of the rule-based temporal extraction, this method is not limited to doctor-patient dialogues. Indeed, with the right adjustments, the method is applicable to a variety of clinical and non-clinical texts.

III. Solution

Our proposed solution combined symptom extraction with a rule-based classifier for temporal extraction.[1] Our first step was to annotate the COVID-Dialogue dataset to manually extract the symptoms and the temporal relationship of those symptoms.

After annotating the data, we had to train or fine-tune a symptom extraction model on the 2010 i2b2/VA dataset. To choose the best model for us to incorporate into our later rule-based classifier for temporal extraction, we looked at some state of the art models for symptom extraction. Due to its ease of its implementation, we used $\text{CharacterBERT}_{\text{medical}}$ (See Boukkouri et al. 2020), the second-highest ranked symptom extraction model of those trained and tested in the 2010 i2b2/VA dataset.

We fine-tuned $\text{CharacterBERT}_{\text{medical}}$, a medical model that was obtained by retraining $\text{CharacterBERT}_{\text{general}}$, on a medical corpus. We specifically chose this version of CharacterBERT in order to potentially transfer knowledge learned over a dataset most similar to that of our dialogs, albeit imperfectly. We fine-tuned $\text{CharacterBERT}_{\text{medical}}$ for 5 epochs using default hyperparameters as specified by the authors.

For the rule-based classifier for temporal extraction, we used the Stanza library from Stanford NLP.[2] We separated the patient dialogues from the COVID-Dialogue dataset into sentences and used the Stanza library to extract the syntax tree dependencies for each of the sentences. In each syntax tree, we located the symptoms extracted from the $\text{CharacterBERT}_{\text{medical}}$ model and found the root verb that the extracted symptom was dependent on. Finally, we determined the tense of the verb, thus allowing us to classify the extracted symptom as a "current" or "past" symptom.

When testing this solution, we noticed that $\text{CharacterBERT}_{\text{medical}}$ was tagging irrelevant stop words as symptoms and worsening our classifier's performance. In an effort to filter out these irrelevant, or edge, stop words, we added a rule to our classifier such that any stop words that were tagged as a symptom would be ignored unless they were directly neighbored by two extracted symptoms that were not stop words. These improvements will be further discussed later.

IV. Experimental Settings

Data

In this study, we used two different datasets in an effort to expose our temporal symptom extractor to a broader range of clinical conversations. To tune the $\text{CharacterBERT}_{\text{medical}}$ model, which was trained on the MIMIC-III dataset, we utilized the Beth Israel subset of the training data used in the 2010 i2b2/VA challenge on concepts, assertions, and relations in clinical text.[3] After fine-tuning the symptom extraction model, we evaluated our temporal symptom extraction model on the COVID-Dialogue dataset.[4]

Data annotation

[1] https://colab.research.google.com/drive/1KIInM9kAZlMAl9skFckE2ANea1Tfnkcg?usp=sharing
[2] https://stanfordnlp.github.io/stanza/
[3] https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3168320/
[4] https://github.com/UCSD-AI4H/COVID-Dialogue

We took 100 entries in the COVID-19 dialogue dataset and split them into 4 groups of 25 dialogues. We each annotated 25 dialogues, marking each symptom word and whether it was a past or current symptom.[5] We wrote a small piece of code which, when given a text and a query, outputs the starting and ending indices of the query, thus making the annotation process much faster.[6] To determine whether a symptom was past or current we predominantly used the tense of the associated verb, while using our own discretion in some cases. For example, "I have had a cough for a while" would extract the symptom 'cough' as current because 'have had' is present perfect, implying it is still ongoing. We agreed on this method beforehand and annotated 25 dialogues individually. On the occasion that the individual annotator could not decide on the temporal label, we discussed the matter as a group and made a collective decision.

Baseline comparisons

Appendix A shows the comparison of baseline models that we considered. One baseline is $\text{CharacterBERT}_{\text{medical}}$. This model is pre-trained, and we performed no fine-tuning on this model. Another baseline was the Clinical Concept Extraction with Contextual Word Embedding (Zhu et al. 2018). This is a bidirectional LSTM-CRF model that was fed embeddings from a contextual word embedding model: ELMo, introduced by Peters et al. 2018. The ELMo embeddings were trained on a corpus consisting of clinical reports from the MIMIC-III dataset and clinically relevant Wikipedia pages. This baseline performs better than our initial $\text{CharacterBERT}_{\text{medical}}$ model fine-tuned. Appendix A also shows our $\text{CharacterBERT}_{\text{medical}}$ model fine-tuned with the removal of edge stop words which showed better performance than the previous baselines mentioned and will be discussed further in the results.

Performance metrics

As a fundamentally binary classification problem, the performance metrics that we analyzed were F1, precision, recall, and support. The best symptom extraction model that we chose showed significantly greater scores in recall, so we concentrated on recall the most in our temporal classifier. In our symptom extraction, recall shows us how many symptoms our fine-tuned model extracted out of all our labelled ground truth symptoms. We also wanted to analyze precision to better understand the false positive rate and our labelling accuracy. As is imperative for any binary classification problem, we wanted to include F1 scores to show the accuracy of our model, a measurement related to both our precision and recall. Finally, the support shows us the class imbalance after temporal extraction.

V. Results

In this section, we show performances for our proposed symptom extraction and

[5] https://docs.google.com/spreadsheets/d/1IhfFJvF0CdCwGZO_xw7a81WTgAdH0B054UzOwW9bcNI/edit?usp=sharing

[6] https://drive.google.com/file/d/1eFanDVhTcZQEEpjuqdHOuuNf_tb2KO1P/view?usp=sharing

temporal information extraction techniques that underlie our overall system.

Appendix A shows class-averaged F1, precision, and recall scores quantifying performances of four distinct models that were considered for the purposes of symptom extraction. Note that the models are labeling each word in a given dialog as either 'problem' or 'O,' ie. *symptom* or *irrelevant word*. All models were evaluated on our COVID-Dialogue dataset. $\text{CharacterBERT}_{\text{medical}}$ (pre-trained), one of our baseline models that was originally trained on data largely disjoint from that which our task deals with (patient-provider dialogues). Due to its consistently poor performances, we believe that it does not generalize well to the clinical domain.

In contrast, Clinical Concept Extraction with Contextual Word Embedding, another of our baseline models (note that this model was used out-of-the-box and without any fine-tuning), appears to exceed the performance of our $\text{CharacterBERT}_{\text{medical}}$ (fine-tuned). Due to time constraints, we sought to focus solely on improving the results of our $\text{CharacterBERT}_{\text{medical}}$ (fine-tuned) model. By manually comparing the predictions of this model across certain samples from the COVID-Dialogue dataset, we tested our hypothesis that exclusion of edge stop words would be helpful towards identifying words that are most relevant to a symptom, leading to the best performances we could achieve across F1, precision, and recall. We believe this shows immense potential for rule-based systems in boosting performances across general symptom extraction procedures, especially procedures trained on more structured information, critically differing from the relatively free-form natural language handled in this study.

Appendix B visualizes the fine-tuning procedure for our $\text{CharacterBERT}_{\text{medical}}$ (pre-trained) model. We selected $\text{CharacterBERT}_{\text{medical}}$ (fine-tuned) as the epoch 4 checkpoint from our training procedure.

Appendix C presents the performances of conducting temporal information extraction upon the predictions of our $\text{CharacterBERT}_{\text{medical}}$ (fine-tuned) model, comparing that when retaining edge stop words neighboring predicted symptoms vs. after removing edge stop words from predicted symptoms. We see marginal improvements from the implemented procedure, particularly across recall of ground truth current symptoms, and we strongly believe that future work (described in the next section) can further improve on these preliminary results.

VI. Limitations and Future Work

Limitations to our work include various colloquial terms of common illnesses in our dataset that may not appear in the 2010 i2b2/VA dataset that we trained on. There were also instances in which the temporal information of a symptom was unclear, such as in the sentence "I was diagnosed with asthma not long ago". Despite the fact that the verb 'was' is past tense, asthma is an ongoing condition and is therefore current. Thus, unclear protocols for annotations of this nature possibly had a negative impact on model performance. Finally, we had limitations in the functionality of our own

rule-based classifier as it handled the dialogue. When recursing from a symptom to a verb phrase like "have had", we should predict "present", but we predict "past" at the moment, likely because the symptom's temporal information relies only on the head of "have had," which is "had". Furthermore, a verb phrase like "I've" currently gets tokenized as "I ve," which is a problem, as we'd prefer it to be tokenized as "I have" for easier temporal extraction from the parse tree.

For future work, we plan on taking multiple steps to both improve and compare our classifier's performance. Firstly, we plan on using Stanford University's CoreNLP in our temporal extraction pipeline.[7] We noticed some syntax dependency errors using Stanza with difficult verbs like past participles, but we were not able to utilize CoreNLP due to time restraints of this project. Since there were many spelling mistakes in the COVID-Dialogue dataset, we also plan on using a spell checker to correct the misspellings. Some corrections would result in different lengths of the words, such as "pneumonia" to "pneumonia", and CharacterBERT uses specific indices of labelled entities in the text to train. This would require us to spell check the entire dataset before labeling occurs. One final possible extension for this project would be a temporal phrase extraction. Symptoms often appear in phrases such as "high AND and AMC levels", and using subtrees in sentence syntax dependency trees would allow us to classify entire phrases that express symptom information as "current" or "past". To compare the accuracy of our rule-based classifier for temporal extraction, we could build in-parallel a BERT model trained on a sequence labeling task. We would label every token in the COVID-Dialogue dataset as *unrelated*, *related to a past symptom*, or *related to a current symptom*. This would combine our pipeline into one model, allowing us to find performance differences between BERT and temporal extractions from sentence syntax trees.

[7] https://stanfordnlp.github.io/CoreNLP/

VII. References

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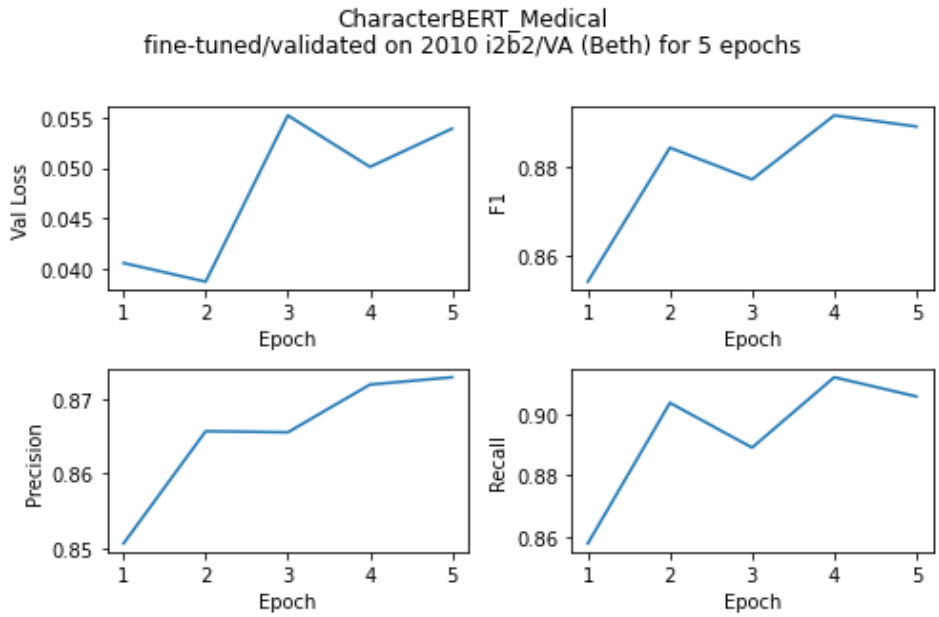
Appendix A

Model	F1	Precision	Recall
CharacterBERT_Medical (pre-trained)	0	0	0
CharacterBERT_Medical (fine-tuned)	0.2495	0.1972	0.5521
CharacterBERT_Medical (fine-tuned) - Edge Stop Words Excluded	**0.3677**	**0.2495**	**0.6986**
Clinical Concept Extraction with Contextual Word Embedding	0.3145	0.2203	0.5493

The above table shows the F1, precision, and recall scores for the baseline model comparisons. Recall was the highest for the final three models. Row 3 shows that our $\text{CharacterBERT}_{\text{medical}}$ fine-tuned model performed better than the other baselines when we excluded stop words that were not between two other non-stop word symptoms.

Appendix B



The above graphs visualize our fine-tuning of the CharacterBERT$_{medical}$ model for 5 epochs. The model we used in this project is from epoch 4 of this training and fine-tuning.

Appendix C

	Precision	Recall	F1	Support
current	0.78	0.73	0.76	291
past	0.61	0.67	0.63	180
accuracy			0.71	471
macro avg	0.69	0.70	0.70	471
weighted avg	0.71	0.71	0.71	471

The above table is the performance evaluating our temporal symptom extraction on the COVID-Dialogue dataset when we did not discard edge stop words.

	Precision	Recall	F1	Support
current	0.78	0.74	0.76	288
past	0.61	0.67	0.64	178
accuracy			0.71	466
macro avg	0.70	0.70	0.70	466
weighted avg	0.72	0.71	0.71	466

The above table is the performance evaluating our temporal symptom extraction on the COVID-Dialogue dataset when we discarded edge stop words.